

Xingyi WU

Curriculum Vitae | September 2026

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1088 Xueyuan Avenue, Shenzhen, China

EDUCATION

Southern University of Science and Technology (SUSTech)

Shenzhen, China

M.S. in Geophysics, GPA: 3.47/4.0

Expected July 2027

Supervisor: Prof. Lei Zhu

Research: Diurnal variations of glyoxal observed from space (GEMS satellite) and interpreted with GEOS-Chem

Nanjing University of Information Science & Technology (NUIST)

Nanjing, China

B.S. in Atmospheric Science, GPA: 3.66/5.0

July 2024

Supervisor: Prof. Jianbing Jin

Research: Time-series analysis and modal attribution of TROPOMI methane observations related to rice cultivation in China

RESEARCH EXPERIENCE

Validation of GEMS glyoxal retrievals over East Asia against TROPOMI, MAX-DOAS, and GEOS-Chem

- Evaluated GEMS glyoxal retrievals using multi-source satellite, ground-based, and model datasets
- Investigated key steps in the satellite Level-2 retrieval algorithm to identify major sources of discrepancies between GEMS and TROPOMI, including differential slant column density, polluted reference area, and uncertainty introduced by a prior profile
- Demonstrated good spatial agreement over land, while identifying GEMS underestimations of 12.3–31.0% relative to TROPOMI and 43.2% relative to MAX-DOAS

Urban diurnal glyoxal sources and sinks revealed by GEMS and GEOS-Chem

- Applied GEOS-Chem to investigate the diurnal variability of urban glyoxal
- Identified rapid chemical production from aromatic compounds and isoprene as major glyoxal sources, with photolysis and aerosol uptake as key loss pathways

Seasonal Methane Concentration Variations Related to Rice Cultivation in China

- Analyzed the spatiotemporal variability of methane over China from 2018 to 2022 using STL decomposition and K-means clustering, identifying four distinct regional modes of seasonal variability associated with rice cultivation

SKILLS

Languages: English (IELTS: 6.5; CET-6: 526)

Programming: Python (main), MATLAB, Fortran, C

Atmospheric Models: GEOS-Chem (main), WRF-Chem, BOXMOX

Computing: Linux, High performance computing (HPC)

Dataset: GEMS and TROPOMI products, EDGAR emission inventories, ASIA-AQ and KORUS-AQ airborne observations, JPL chemical kinetics, Google Earth Engine

PRESENTATION

1. Oral presentation, *Investigating Glyoxal (CHOCHO) Sources and Sinks in the Pearl River Delta through GEMS Satellite Observations and BOXMOX Model Simulations*, the AOGS 2025 Annual Meeting, Singapore, Singapore, Jul 30, 2025
2. Oral presentation, *Evaluation and Interpretation of GEMS Glyoxal Products*, the 17th GEMS International Workshop, Seoul, Korea, Sep 10, 2026
3. Oral presentation, *Satellite Observations and Chemical Simulations of Diurnal Glyoxal Variations in Urban Areas*, the 6th

4. Oral presentation, *Multi-scale Observations and Chemical Simulations of Urban Diurnal Glyoxal Variations*, the 13th National Conference on Environmental Chemistry (NCEC 2025), Jiangmen, Guangdong, Nov 30, 2025

PUBLICATIONS

1. **Wu, X.**, Zhu, L. *, Zhang, A., Li, X., Fu, W., Kwon, H.-A., Park, R. J., Ha, E. S., Lee, G. T., Min, K.-E., Liu, S., Li, Y., Zhang, X., Zhang, P., Zhang, J., Huang, Y., Zhang, L., Li, J., Luan, X., Fu, T.-M., Yang, X., Shen, H., Ye, J., and Wang, C.: Evaluation of GEMS glyoxal retrievals with TROPOMI product, MAX-DOAS observations, and GEOS-Chem simulations, submitted to *Earth and Space Science*.
2. **Wu, X.**, Zhu, L. *, Zhang, A., Fu, T.-M., Zhang, X., Fu, W., Kwon, H.-A., Park, R. J., Ha, E. S., Lee, G. T., Min, K.-E., Liu, S., Li, Y., Li, X., Li, Y., Zhang P., Zhang J., Huang Y., Zhang L., Li J., Luan X., Yang, X., Shen, H., Ye, J., and Wang, C.: Distinct Diurnal Regimes of Glyoxal over East Asian Cities Driven by Production-Loss Competition, *in prep*.
3. **Wu, X.**, Jin J. *, Xia J., Fang, L., and Liao H.: Seasonal Methane Concentration Variations Related to Rice Cultivation in China, *Chinese Journal of Atmospheric Sciences*, 50(2): 463–477, <https://doi.org/10.3878/j.issn.1006-9895.2504.24111>, 2026. (*in Chinese*)
4. Li, X., Liu, S. *, Zhu, L. *, Zhang, P., Zhang, X., Li, Y., **Wu, X.**, Zhang, J., Huang, Y., Luan, X., Zhang, L., Yang, X., Shen, H., Ye, J., Wang, C., and Fu, T.-M.: Vertical Redistribution of Ozone Precursor Proxies during Typhoon Peripheral Periods: Insights from MAX-DOAS and GEOS-Chem, *Journal of Geophysical Research-Atmospheres*, 131, e2026JD046455, <https://doi.org/10.1029/2026JD046455>, 2026.
5. Li, Y., Zhu, L. *, Li, J., Chen, Y., Zhang, L., Montzka, S., Vimont, I., Crotwell, Molly., Liu, S., Li, X., Fu, W., Zhang, P., Zhang, X., **Wu, X.**, Zhang, J., Huang, Y., Luan, X., Shen, H., Ye, J., Wang, C., Yang, X., and Fu, T.-M.: Assessing the Accuracy of Global HFC-134a Satellite Observations from Atmospheric Chemistry Experiment Fourier Transform Spectrometer (ACE-FTS) via Aircraft, Balloon, and Model Comparisons, *Environmental Science & Technology*, 60 (1): 959–972, <https://doi.org/10.1021/acs.est.5c13191>, 2025.
6. Zhang, P., Zhu, L. *, Zhai, S., Liu, S., Bates, K., Fu, W., Li, X., Chen, Y., **Wu, X.**, Li, Y., Li, J., Zhang, X., Zhang, J., Huang, Y., Yang, X., Fu, T.-M., Shen, H., Ye, J., and Wang, C.: Wintertime Peroxyacetyl Nitrate (PAN) Hotspot over China Observed from Space: Interpretation of CrIS Results Using Updated GEOS-Chem, *Environmental Science & Technology Letters*, 13 (1): 76–82, <https://doi.org/10.1021/acs.estlett.5c00950>, 2025.
7. Li, Y., Zhu, L. *, Li, J., Chen, Y., Western, L. M., Young, D., Mühle, J., Weiss, R. F., Krummel, P. B., Lunder, C. R., Liu, S., Li, X., Fu, W., Zhang, P., Zhang, X., Zhang, J., **Wu, X.**, Huang Y., Shen, H., Ye, J., Wang, C., Fu, T.-M., and Yang, X.: Developing the chemistry module for 27 fluorinated greenhouse gases (F-gases): Reactions, emissions, and implementation in GEOS-Chem, *Atmospheric Environment*, 121604, <https://doi.org/10.1016/j.atmosenv.2025.121604>, 2025.
8. Zhang, X., Zhu, L. *, Ye, C., Kim, J., Lee, H., Park, J., Jung, Y., Hong, H., Fu, W., Li, X., Chen, Y., **Wu, X.**, Li, Y., Li, J., Zhang, P., Yan, Z., Zhang, J., and Liu, S.: Tropospheric NO₂ column over Tibet Plateau according to GEMS: Spatial, seasonal, and diurnal variations, *Remote Sensing*, 17(10), 1690, <https://doi.org/10.3390/rs17101690>, 2025.
9. Fu, W., Zhu, L. *, Kwon, H.-A., Park, R. J., Lee, G. T., De Smedt, I., Liu, S., Li, X., Chen, Y., Pu, D., Li, J., Zuo, X., Zhang, P., Li, Y., Yan, Z., Zhang, X., Zhang, J., **Wu, X.**, Shen, H., Ye, J., Wang, C., Fu, T.-M., and Yang, X.: Evaluating GEMS HCHO retrievals with TROPOMI product, Pandora observations, and GEOS-Chem simulations, *Earth and Space Science*, 12, e2024EA003894. <https://doi.org/10.1029/2024EA003894>, 2025.

AWARDS AND HONORS

- First-Class Academic Scholarship for Graduate Students, SUSTech, 2025
- International Conference Grant, 10000 RMB, SUSTech, 2025
- Outstanding Postgraduate Teaching Assistant, SUSTech, 2025
- Outstanding Postgraduate Students, SUSTech, 2024–2025
- Outstanding Undergraduate Graduates, NUIST, 2024